Module 4: Bayesian Inference

INFO 521 · Machine Learning Foundations · Week 4 · module overview

What should we believe about a parameter, before and after seeing data?

Module 3 gave you a single best estimate and an error bar around it. This module replaces the point estimate with a distribution you can update. When the prior and the likelihood are a conjugate pair, that update is closed form: no optimisation, no simulation, just arithmetic on the parameters.

What the lectures do

Lecture A: Priors, Posteriors, and the Beta-Binomial. Bayes' rule term by term, hypertension prevalence as the running question, the Beta prior, conjugacy as addition, and how quickly the prior stops mattering.

Lecture B: The Conjugate Gaussian Posterior. The same idea for the regression weights, the posterior contracting as data arrives, ridge reappearing as a particular prior, and the predictive band for a new patient.

Where this sits in the course

Ridge regression showed up in Module 2 as a penalty you add because it works. Here it reappears as the consequence of a Gaussian prior, which is the sort of thing that happens repeatedly once you take the probability model seriously. Module 5 then breaks the conjugacy that makes all of this easy.

Week 4 at a glance

- Homework Unit 4, exact and conjugate Bayesian inference.
- Checkpoint quiz 1.3 on the conjugate posterior. Passing it opens Milestone 1.3.
- Milestone 1.2, maximum likelihood, is due this week.
- Activity loop: the week03 Bayesian-updating explorer.
- Discussion: how the posterior contracts.
- Reading: PML sections 4.6 and 11.7; Think Bayes Chapters 2 and 4.

Watch out for

The MAP is not the posterior. It is the single highest point of it. Reporting the MAP alone throws away the spread, which was the reason for going Bayesian in the first place.

Conjugacy is a convenience, not a law. It holds for the pairs in this module and fails as soon as the likelihood changes shape. Module 5 is what happens next.